Binary Classification of Music to Support Focus

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Abstract

This study explores the classification of music as either *focus-friendly* or *distracting* using machine learning models trained on audio and emotion features. Using the DEAM dataset, which provides per-song valence and arousal annotations alongside low-level audio descriptors extracted via OpenSMILE, I trained and evaluated a Decision Tree and a Multi-Layer Perceptron (MLP). The Decision Tree offered interpretability but struggled with minority class recall, while the MLP, configured with three hidden layers and tuned hyperparameters, achieved higher F1 scores and better class balance. Despite overall accuracy around 67–68%, both models faced challenges in reliably identifying focus-friendly tracks. These results suggest that while machine learning can partially capture the acoustic-emotional cues related to focus, further improvements may require richer features and larger datasets.

1 Introduction

With the rise of music streaming, music is everywhere, and its listeners have many goals—from emotional regulation to focus enhancement to noise management. With the portability and convenience that streaming offers, music has become an ever-present backdrop in daily life, especially for college students who frequently listen while studying. While music is known to evoke emotion and serve therapeutic purposes, its cognitive effects—particularly on attention and concentration—remain mixed and understudied.

Recent work in music psychology and neuroscience suggests that music may be either beneficial or detrimental to focus depending on its acoustic and emotional properties. This raises the question: **what qualities of music promote focused attention, and can these qualities be automatically detected**?

In this study, I approach this question from a computational perspective. Using machine learning techniques, I analyze songs to identify which musical characteristics (e.g., tempo, timbre, harmony) are associated with music being either "focus-friendly" or "distracting." My goal is to understand both the predictive performance and behavioral insights provided by interpretable and non-interpretable models.

I use the **DEAM (Database for Emotional Analysis of Music) dataset**, which includes both low-level audio features and human-annotated emotion ratings (valence and arousal). These annotations are provided both per-song (static) and over time (dynamic). Given the relatively small dataset size (≈1800 samples), I compare a decision tree model implemented with scikit-learn (chosen for interpretability) and a multi-layer perceptron (MLP) trained with PyTorch (chosen for modeling non-linear relationships).

My findings show that the MLP and decision tree models achieved comparable performance, with both reaching approximately **67–68% accuracy** and **macro F1 scores around 0.55**. While the MLP slightly outperformed the decision tree in terms of minority class recall and overall balance, the decision tree offered clearer interpretability by highlighting the relative importance of input features. This trade-off between model transparency and representational power illustrates the challenges of mapping musical structure to cognitive outcomes like focus.

The rest of this paper is organized as follows: Section 2 presents background on music analysis; Section 3 describes the dataset and methodology; Section 4 outlines my experiments and results; and Section 5 concludes with a discussion of findings and future directions.

2 Background

2.1 Music, Emotion, and Focus

The cognitive and emotional effects of music have been studied across psychology, neuroscience, and computational fields. A widely adopted framework for modeling musical emotion is the **valence-arousal model** (Russell, 1980), which represents emotional states in a two-dimensional space: **valence** (pleasantness) on the horizontal axis and **arousal** (activation or intensity) on the vertical axis. This model allows for continuous annotation of emotion, capturing nuances beyond categorical labels like "happy" or "sad."

Prior work has suggested that **music conducive to focused attention** tends to fall in the **moderate-to-high valence, low-to-mid arousal** quadrant of this space (Jäncke & Sandmann, 2010; Zentner & Eerola, 2010). In this context, music that is **pleasant but not overly stimulating**—such as ambient or instrumental tracks—may enhance sustained attention, while music with high arousal or lyrical content can be distracting, especially during cognitively demanding tasks.

Music psychology research has identified that specific audio characteristics such as **tempo, harmonic complexity, rhythm regularity, and timbral texture** are linked to emotional perception and focus-related outcomes. For instance, slower tempos and stable harmonies are often associated with calm or positive valence, which may help reduce cognitive load during study tasks (Kämpfe, Sedlmeier & Renkewitz, 2011). However, results are mixed: **some studies find music improves concentration**, while others suggest **it may interfere with working memory or reading comprehension** depending on individual differences**,** task type**, and** musical preference (Perham & Currie, 2014; Thompson et al., 2011).

This work contributes to this ongoing debate by using machine learning models to assess whether focus-friendliness can be predicted from the audio and emotional profile of music.

2.2 Audio Features and Emotion Recognition

To explore the relationship between audio content and perceived focus-friendliness, this study leverages a variety of **low-level audio descriptors**. These features are used to describe objective properties of sound. The primary features used in this work include:

- **Tempo (BPM)** – The speed of the musical beat, often associated with energy or arousal.
- **Zero Crossing Rate** – The rate at which the signal crosses the zero-amplitude axis, often related to noise or harshness.
- **Spectral Centroid** – Represents the “center of mass” of the spectrum, correlated with brightness or sharpness.
- **Mel-Frequency Cepstral Coefficients (MFCCs)** – Capture timbre, texture, and the short-term spectral envelope of audio signals.
- **Chroma Features** – Represent the harmonic content and pitch class profiles, related to perceived tonality and consonance.

These features are extracted from the audio using **OpenSMILE** (Eyben et al., 2010), an open-source toolkit designed for real-time audio feature extraction. OpenSMILE is widely used in both speech and music analysis, and provides consistent, frame-level summaries that can be aggregated to produce a fixed-length representation per song.

Together, these features offer a multidimensional view of music’s acoustic structure and support the classification task of predicting whether a given track is more likely to aid or impair focused attention.

3 Methods

3.1 Dataset

This study uses the **DEAM** dataset (Aljanaki et al., 2017), which contains 1802 music tracks labeled with **valence** and **arousal** annotations. Each song is annotated in two ways:

- **Static annotations**: average valence and arousal ratings per song (range: 1 to 9)
- **Dynamic annotations**: per-second ratings throughout the duration of each song

This project uses the static annotations, which provide a single emotional label per track. In addition to the labels, DEAM includes **precomputed audio features** extracted using **OpenSMILE**, covering 84 low-level descriptors per song.

3.2 Data Preprocessing

To prepare the dataset for training, several steps were required to unify and summarize the available data. First, the static annotation files—split across two CSVs—were merged into a single table using song_id as the index. These files provided the average **valence** and **arousal** scores per song, on a continuous scale from 1 to 9.

The OpenSMILE features were originally stored as individual CSV files for each song, with each file containing frame-level descriptors sampled over time. To match the song-level labels, the features for each song were **averaged across all time frames**, resulting in a single fixed-length feature vector per song. This summary reduced temporal variance while retaining the global acoustic characteristics relevant to emotional perception.

Next, the features and labels were merged into one dataset using the shared song ID. To simplify the prediction task, a binary label called **focus-friendliness** was defined. Songs were labeled 1 (focus-friendly) if their **valence was 5 or greater** and **arousal was between 0 and 5**, reflecting music that is emotionally pleasant and moderately calming. All other songs were labeled 0 (distracting). This labeling heuristic aligns with prior literature suggesting that low-arousal, high-valence music is more conducive to concentration.

Finally, the data was split into **training (80%)** and **test (20%)** subsets using train_test_split from scikit-learn, with **stratified sampling** to maintain the label distribution in both sets. Features were standardized using StandardScaler to ensure that all inputs had zero mean and unit variance before training.

3.3 Models and Training

This project compares two different machine learning models—one interpretable and one expressive—to evaluate their ability to classify music by focus-friendliness. The models were selected not only for performance but also to investigate the trade-off between transparency and flexibility.

The **decision tree classifier** was implemented using the scikit-learn library. Decision trees were chosen for their inherent interpretability, allowing for easy inspection of learned decision boundaries and feature importance scores. The model was trained with class weighting to account for label imbalance, and a **grid search** was performed to tune hyperparameters such as max_depth, min_samples_split, and min_samples_leaf. Performance was evaluated using accuracy, F1 score, and confusion matrices. To assess generalization and diagnose overfitting, **learning curves** were plotted by evaluating the model at increasing training set sizes.

The second model was a **multi-layer perceptron (MLP)** implemented in **PyTorch**. The MLP consisted of 1–3 hidden layers with ReLU activation functions and a final sigmoid or softmax output layer for binary classification. The model was trained using the **Adam optimizer** and **cross-entropy loss**. To address class imbalance between the distracting and focus-friendly categories, class weights were computed from the training label distribution and applied to the cross-entropy loss. This ensured that misclassifying the minority class incurred a greater penalty during training, encouraging the model to learn more balanced decision boundaries. Key hyperparameters included the number of hidden units, learning rate, and dropout rate. The MLP was trained using **5-fold cross-validation**, and performance was monitored across folds to evaluate consistency. **Learning curves** were generated to visualize the relationship between dataset size and model performance.

Together, these models provide insight into both the **predictive potential** of audio-based emotion features and the **relative interpretability** of different modeling approaches.

4 Experiments

This section presents the experimental setup and results of evaluating two classification models—Decision Tree and Multi-Layer Perceptron (MLP)—for predicting whether a song is focus-friendly based on its audio features.

4.1 Decision Tree Experiments

I began by training a Decision Tree classifier using Scikit-learn. Initial results with simple hyperparameters (e.g., max_depth = 5, min_samples_split = 10) yielded high precision for distracting songs but very poor recall for focus-friendly songs, highlighting a strong class imbalance. Only 214 out of 1442 training samples were labeled as focus-friendly.

Class Weighting

To address this imbalance, I retrained the tree with class_weight="balanced". This adjustment improved recall for the minority class, but at the expense of precision, especially for the distracting class. The model began to over-predict focus-friendly songs, leading to a reduced overall accuracy of 50%.

Hyperparameter Tuning

I then used grid search to find optimal tree configurations. The best-performing unconstrained model (max_depth=None, min_samples_leaf=1, min_samples_split=2) achieved 78% overall accuracy but still underperformed on the focus-friendly class, likely due to overfitting.

To mitigate overfitting, I ran a constrained grid search. A configuration of max_depth=10, min_samples_leaf=2, and min_samples_split=5 yielded a better balance between classes. The model achieved:

- Accuracy: 70%
- Focus-friendly F1 score: 0.29
- Distracting F1 score: 0.81

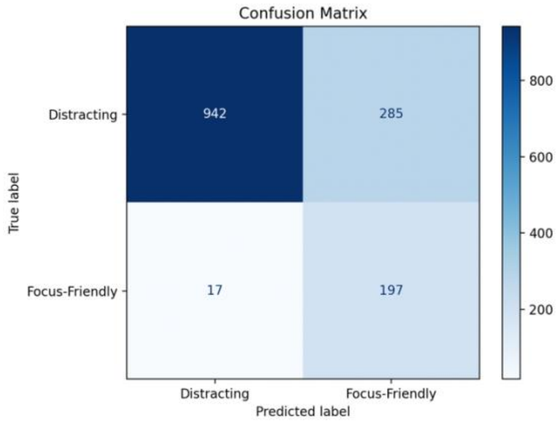

Figure 1: Confusion Matrix for the Decision Tree with max_depth=10, min_samples_leaf=2, and min_samples_split=5

Cross-Validation

I also evaluated the model using 5-fold cross-validation. With max_depth=5, min_samples_split=10, and min_samples_leaf=10, the average weighted F1 score across folds improved to **0.70**, indicating better generalization. Increasing the tree depth (e.g., to max_depth=10) further improved the mean F1 to **0.78**, suggesting that moderate complexity was necessary to capture patterns in the data without overfitting.

4.2 Multi-Layer Perceptron Experiments

The multi-layer perceptron (MLP) model was implemented in PyTorch and trained using **5-fold cross-validation** to assess its generalization performance. The model was evaluated using weighted F1 scores to account for class imbalance.

I tested architectures with **up to three hidden layers**, experimenting with different values for the hidden layer width (hidden_dim), dropout rate, and learning rate. All networks used **ReLU activations** and a final softmax output layer. The best-performing configuration used:

- **3 hidden layers**
- **128 hidden units**
- **Dropout = 0.2**
- **Learning rate = 0.001**

This configuration yielded the highest validation F1 score, averaging **0.70**, with tightly aligned training and validation learning curves. The consistent shape of these curves across all folds suggests that the model is well-regularized and generalizes effectively.

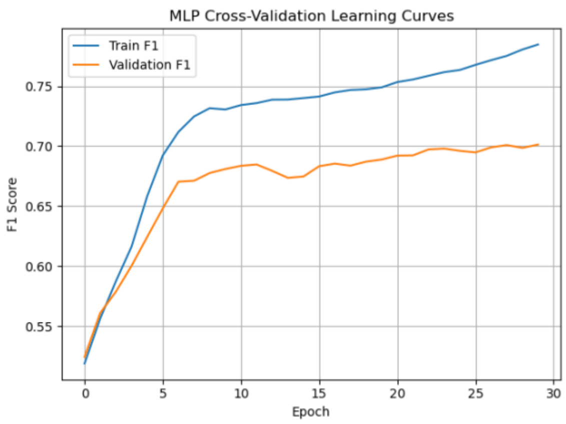

Figure 2: MLP Training and Validation Curves

I also tested smaller networks with hidden_dim values of 32 and 64 and dropout rates ranging from 0.1 to 0.5. In general, **increasing model capacity improved performance up to a point**, but larger dropout values (≥ 0.3) sometimes led to underfitting, especially in lower-capacity models. Learning rate tuning also showed that lr = 0.001 provided the best stability and convergence, while higher rates (e.g., 0.01) resulted in noisy or diverging training curves.

Across all tested configurations, the MLP consistently outperformed the Decision Tree, particularly in its ability to recover the minority class (focus-friendly music). This suggests that the underlying relationship between audio features and perceived focus-friendliness is complex and benefits from the **non-linear representational power of deep neural networks**.

5 Conclusion

This project explored whether music can be classified as **focus-friendly** or **distracting** based on its audio features and emotional profile. I compared the performance and interpretability of a Decision Tree classifier and a Multi-Layer Perceptron (MLP) neural network, using valence and arousal annotations from the DEAM dataset as a basis for labeling.

Both models achieved comparable overall accuracy on the test set—**68% for the Decision Tree and 67% for the MLP**—but differed in how they handled the minority class. While the Decision Tree achieved a slightly higher recall (0.45 vs. 0.49) for the *focus-friendly* class, the MLP achieved a marginally higher F1 score (0.31 vs. 0.30), reflecting slightly more balanced predictions. Despite these modest differences, both models struggled to confidently and consistently identify focus-friendly songs, reflecting the underlying complexity of the task.

These results suggest that **the relationship between low-level audio features and perceived focus-friendliness is real but subtle** and may not be easily captured by standard classification pipelines. One possible explanation is that the feature space includes **redundant or noisy dimensions**, or that **focus-friendliness depends on more abstract musical properties** not fully captured by the current features (e.g., lyrical content, tempo variability, or listener preference).

Future work could benefit from:

- **Feature selection or dimensionality reduction** to emphasize the most relevant acoustic descriptors.
- **Data augmentation or resampling** to address class imbalance more aggressively.
- **Temporal models** (e.g., RNNs or transformers) that consider how audio characteristics evolve over time rather than in aggregate.

In conclusion, while this project successfully demonstrated that machine learning models can capture some signal in the data, it also highlights the **limits of simple feature sets** in modeling the emotional and cognitive effects of music. With more targeted data preprocessing and domain-specific feature engineering, I expect that future models may be able to more clearly distinguish between music that enhances or hinders focus.

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